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Complex Training Revisited: A Review of its Current Status as a Viable Training Approach

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summary

Complex training has been proposed as a training method to increase power. This article reviews the current literature on complex training, its effect on performance, and possible physiological explanations for its effects.

Power is determined from work per unit of time and considered to be a fundamental aspect of successful athletic performance, especially in sports that require speed, agility, and explosive actions (7, 17, 27, 39). Consequently the optimal training techniques to maximize power and the transfer of power to athletic performance have received considerable attention from researchers and sport conditioning coaches. The three basic training approaches that have been used to develop power include traditional weight training with heavy loads (80–90% of 1 repetition maximum [RM]), plyometric exercises incorporating acceleration and deceleration of body weight, and dynamic weight training in which the athlete moves a load of 30–50% 1RM as fast as possible (40). Combination training has also been used in a number of different ways that involves the use of both heavy and light loads in some form of alternating manner but not within the same training session (13) or as part of a periodized training regimen (19).

More recently, a number of researchers and practitioners have advocated the use of complex training (CT) techniques, a term credited to Verkhoshansky et al. (39). Although the term has been used to describe slightly different approaches to training, CT generally involves the execution of a resistance-training exercise using a heavy load (1–5RM) followed relatively quickly by the execution of a biomechanically similar plyometric exercise (7, 17, 41). An example would include 5 repetitions of a front squat using a 5RM load followed by 6–8 vertical jumps or depth jumps. The coupling of a strength training exercise with a plyometric exercise is often referred to as a complex pair. The pair of exercises is then usually repeated for a number of sets. Despite the popularity and advocacy of CT as a method for enhancing explosive power (7, 17, 41), there is still a lack of scientific support for this approach. Ebben and Watts (10) conducted a review of the concept and research related to CT and concluded that it deserved objective consideration and careful analysis to determine its benefits and identify the underlying mechanisms that would contribute to its efficacy. More recently, Ebben (9) revisited the concept of CT and concluded that it is at least as effective, and possibly superior, as other forms of combined strength and plyometric training. However, as this review will indicate, there are probably still more questions than answers in regard to the effectiveness of complex training in developing athletic power. The purpose of this review is to examine the physiological rationale for CT and the research that has examined the acute and chronic effects on enhancing performance.

The Physiological Rationale Behind CT

The premise on which complex training is based assumes that the explosive capability of muscle is enhanced after it has just been subjected to maximal or near-maximal contractions. This phenomenon has been referred to as postactivation potentiation (PAP) (18, 29, 33).

The potentiated state of muscle is considered to have an acute effect on enhancing its acute performance capability, which repeated over time, as in a training program, is expected to produce superior chronic adaptations compared with other training approaches (7, 17).

Two theories or mechanisms have been proposed to explain the potentiated state of muscle after maximal or near-maximal stimulation. One theory proposes that the prestimulation enhances motor-neuron pool excitability, as evidenced by a potentiated reflex response (17, 37). The increased neural activation may occur through recruitment of more motor units (MU), better MU synchronization, a decrease in presynaptic inhibition, or greater central input to the motor neuron (1, 2).

Phosphorylation of the myosin light chain (MLC) has been considered an alternative mechanism that is responsible for PAP after a stimulus to the muscle (29, 31, 33). Muscle stimulation causes an increase in sarcoplasmic Ca^{2+} that activates MLC kinase (MLCK) (31, 33, 36). MLCK is responsible for making more ATP available at the actin-myosin complex that, in turn, increases the rate of actin-myosin crossbridging. The phosphorylation of the MLC also renders the actin-myosin interaction more sensitive to Ca^{2+} released from the sarcoplasmic reticulum (35). However, Tubman et al. (38) examined MLC phosphorylation and posttetanic potentiation and concluded that it was not the only mechanism contributing to PAP. It is possible that PAP is the result of interactions between neural and muscular mechanisms that are not well understood at this time.

Evidence for the Existence of PAP

Two approaches have been taken in research to examine the existence of PAP. One approach has examined the twitch contractile properties of muscle, and the other athletic performance that uses explosive actions.

Twitch Contractile Properties

A twitch is a brief contraction of a muscle in response to short (<1 ms) electrical stimulation of a nerve (28). The twitch contractile properties are measured before and after some type of muscle activity, referred to as the contractile history, usually in the form of a series of submaximal evoked twitches (26), evoked tetanic contraction (28), or sustained maximal voluntary contractions (15). Regardless of the type of contractile history of the muscle, the studies examining the contractile properties of muscle have consistently shown a potentiated response referred to as twitch potentiation (TP). Specifically, these studies have demonstrated increases in peak twitch force and the rate of force development, as well as a decrease in relaxation time. TP is a well-established and reproducible phenomenon that, in part, has led to other researchers and practitioners exploring its application for athletic performance. The second approach in examining the phenomenon of PAP has involved activities requiring explosive actions, which have greater potential application to enhancing training or athletic performance. Unfortunately, the evidence of its efficacy in enhancing performance is less clear.

PAP and Functional Performance

Acute Studies

Most studies that have explored the effects of a heavy preload exercise on subsequent explosive movements have examined acute responses, often implying chronic neuromuscular adaptations would occur if the protocol was used in a training regimen. The results of these studies have been equivocal, due, in part, to the number of factors that must be considered in the application of PAP to performance measures, such as the magnitude and mode of the preload activity, the length of the time period between the preload and outcome measures, and the training status of the participants.

The majority of studies that have been conducted have used maximal or near maximal contractions to provide the preload stimulus, either in the form of isometric or dynamic muscle actions. Isometric muscle actions have an advantage in comparison to dynamic actions in that they are more feasible to use in an athlete's warm-up if the intent is to affect the immediate performance (14). Güllich and Schmidtbleicher (17) were among the first researchers to suggest that subjecting a muscle to maximal or near-maximal efforts would enhance subsequent performance, especially explosive actions or mechanical power, with probable chronic adaptations. They preloaded participants by having them perform 3–5 maximal voluntary isometric contractions (MVICs) using unilateral leg press and bench press positions. After the 5-second MVICs, the participants pushed a guided barbell as fast as possible for 5 repetitions, with a 30-second rest interval between repetitions, or performed 8 vertical jumps on a dynamometric force platform. They concluded that the use of a few MVICs was sufficient to increase explosive force in the upper and lower extremities and could be used to enhance performance and training. It should be noted that this study is one of the primary studies cited in support of the concept of CT. However, it is only available as a translation from the original article written in German and lacks clarity in regard to the design and methodology.

A more recent study examined the effect of different durations of the MVIC in an attempt to identify the optimal time period the muscle should be subjected to the maximal effort. Three repeat actions of 3-second isometric knee extensions were found to be more effective than 3 repeats of 5-second actions but only for movements that required some form of stretch-shortening cycle response (14). However, other studies using different durations (2.5–10 seconds) for the MVICs have failed to find any enhancement in subsequent performance (23, 32, 33). It would appear that the efficacy of MVICs and the optimal time period for the actions on subsequent explosive movements is not clearly established at this time.

It is possible the equivocal findings of studies using isometric actions as the preload activity is partly attributable to differences in the mode of muscle action between the preload stimulus and the explosive movement. A number of studies have used dynamic muscle actions as the preload exercise, using loads of 3–5RM, before performing the explosive movement. Young et al. (41) found a 1-cm increase in the mean of five vertical jumps after a set of half squats with a 5RM load. A 3RM half-squat load has also been found to produce an improvement in force generation but only after the participants were divided into high- and low-strength groups (8). However, other studies using a 5RM load did not find any enhancement in the subsequent explosive performance (11, 20, 22, 34).

Although early investigators stressed the importance of a maximal or near maximal preload, a few studies have used lighter loads and found some evidence of potentiation. The effects of 5 different warm up protocols on horizontal jump performance were investigated using a variety of preloads (30). A significant improvement only occurred after the warm-up protocol that included 4 power snatches at 75–85% of a 4RM load. More recently, a 4.5% increase in power output, as measured by a bench press throw with a resistance of 50 kg performed in a specially designed plyometric apparatus, was found after 6 repetitions at 65% of a 6RM load (4). A preperformance protocol of 5 sets of half squats of 2 repetitions each at 20, 40, 60, 80, and 90% of 1RM half-squat load produced a 2.39% increase in vertical jump height (16). However, it is not possible from this study to identify which load produced the potentiated response or if it was the combination of light and heavy loads that affected the outcome. It may be that maximal or near maximal efforts are not required to elicit a potentiated response or some combination of loads may provide the optimal stimulus. However, at this time, the relationship between the magnitude of the preload and explosive performance needs to be more clearly defined.

One of the problems associated with PAP is that the muscle may be fatigued from the heavy preload stimulus that would mask any potential potentiated effect (15, 25, 31). It is clear that potentiation and fatigue can coexist (31). Therefore, it is important to identify the time when the muscle has partially recovered from fatigue but is still potentiated. The time period between the preload stimulus and the execution of a subsequent activity has varied between studies (15 seconds–18.5 minutes). Although difficult to interpret from the translated paper, Güllich and Schmidtbleicher (17) found that the potentiated effect appeared 3–5 minutes: 20 seconds after 3 MVICs. Most of the subsequent studies have used a 3–4 minute rest period between the preload activity and the performance measure, apparently based on the findings of Güllich and Schmidtbleicher (17). Only one study has directly manipulated the length of the rest period between the preload and the explosive activity (21). Rest periods of 1, 2, 3, and 4 minutes were chosen to determine if there was an optimal time for performing plyometrics after a strength-training exercise (5RM squat). No significant improvement in performance occurred after any of the rest periods. The authors concluded that CT does not appear to significantly enhance jumping performance and actually decreases when performed immediately after the resistance exercise. They did imply that it might require more than 4 minutes of recovery for performance to be enhanced. The optimal recovery time between the complex pair of exercises would appear an important factor in optimizing performance but has yet to be identified. It may also differ between participants and, in a practical setting, need to be individually determined and applied (17).

Training status has been suggested to have an effect on the ability of individuals to use the potentiated effects from a heavy preload activity on subsequent explosive performance (8, 12, 17, 41). A significant correlation was found between jump improvement after preload half-squats and 5RM squat strength, which led to the conclusion that stronger individuals may be better able to use the potentiated effect. Although no initial enhancement in performance was found after a 3RM half-squat, there was an overall improvement in force generation for participants after they were classified as high strength or low strength (8). Participants defined as athletically trained, as well as participants competing in power sports, have also been found to be more able to capitalize on PAP compared with groups described as recreationally trained or physical education students (6, 17). However, other authors have concluded that there was no ergogenic effect from a 5RM squat stimulus, even when the participants were divided into high- and low-strength groups (21). At this time, the relationship between strength and the ability to use PAP must be considered tenuous because the studies that have been described were retrospective and did not use strength training as an intervening variable. Neither has an optimal level of strength been identified. In addition, the studies that have suggested strength as a factor in potentiation have used absolute rather than relative values, even when using performance measures, such as the vertical jump, that involve propulsion of body weight. It would seem beneficial to use relative and absolute strength measures to more clearly define the relationship between strength and potentiated performance.

Most of the studies examining the concept of CT protocols have used a single set of the preload activity. The application of PAP to CT for long-term neuromuscular adaptations generally involves the performance of multiple sets of the complex pairs (7). Only two studies have examined the efficacy of applying the principles of PAP to performance after 3 sets of the complex pair. One study attempted to replicate the structure of a CT protocol by investigating the effect of a 7-second MVIC in the half-squat position over 3 consecutive sets on power output measures recorded from a force platform during 5 countermovement jumps (CMJs) (32). No potentiated effect was found in any of the sets, and one measure actually decreased from the first to the last set. The effects of a 3RM load was monitored for the mean of 4 CMJs over 3 consecutive sets (8). From the initial analysis, there was no significant enhancement in performance or for any of the kinetic measures in any of the sets. The participants were then divided into high- and low-strength groups based on the median of their predicted 1RM scores. It was found that the stronger participants had more force improvement (the mean for the 3 sets) after the 3RM half-squat preload. At this time, it appears the ability to produce and retain potentiated effects over a number of sets is not well defined and needs further investigation.

It is possible that participants need repeated exposure to the CT protocol to learn to use the potentiated effects produced by heavy preload activities. However, participants who were exposed to 3 training sessions performed at least 48 hours apart, involving a 5RM half-squat preload before performing 8 CMJs and horizontal jumps, failed to show any improvement in either of the performance measures (34). It is possible that more exposure to the CT-training protocol may be required to benefit from PAP in a practical setting.

Chronic Studies

Only one study was found that directly investigated the efficacy of a CT program in producing long-term neuromuscular adaptations. The effects of a CT program were compared with a combined training program that involved a combination of strength training and plyometrics but not performed as a complex pair, with the plyometric exercise performed immediately after the resistance exercise (5). At the end of a 7-week training period, both groups had increased on 8 of 10 dependent measures, but only the CT group demonstrated an increase in vertical jump. Neither group showed an increase in upper-body power, as measured by a medicine ball throw. However, the study is only available in abstract form, so it is not possible to identify the weight exercises that were used, the magnitude of the loads, the rest intervals, or the specific methodology used in the training programs. Two other studies (3, 24) have also been cited in previous reviews on CT (9, 10) as investigating the efficacy of training protocols that appear to use the principles of CT on chronic adaptations to explosive performance, although neither actually used the term. Because they did not use the actual CT protocols based on the temporal proximity between the strength training and plyometric exercises, they have not been incorporated into this review.

Conclusion

The concept of PAP has been applied to enhance acute, voluntary explosive performance as well as to provide a rationale for CT. There has been lack of a systematic approach to the development and design of the studies investigating the application of PAP to measures of explosive performance that has, in part, contributed to the equivocal findings. To assess the validity of CT as a training strategy, there is a need to conduct more chronic studies that compare the effectiveness of CT to other training regimens aimed at enhancing muscular power. ♦

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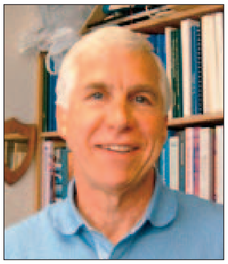

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